

Introduction to Exercise Physiology How Your Body Responds to Movement

Joanne Donoghue, PhD. ACSM-CEP

Director of Clinical Research

Professor

Department of Osteopathic Manipulative Medicine

New York Institute of Technology College of

Osteopathic Medicine

jdonoghu@nyit.edu



Session Objectives

- Describe the three primary energy systems that fuel activity./Learning Objective 1
- Explain how VO_2 , METs, and RER relate to exercise intensity fuel utilization and weight loss./Learning Objective 2.
- Describe the cardiovascular and respiratory responses to exercise and explain how these responses guide exercise prescription in individuals with chronic disease./Learning Objective 3
- Understand relative and absolute contraindications to exercise testing ./ Learning Objective 4.
- Describe how ventilatory and cardiac output responses are influenced by various physiological and environmental factors, including exercise intensity, temperature, and altitude./Learning Objective 5

Lecture Outline: Intro to Exercise Physiology

Topic	Time (min)
Introduction to Exercise Physiology (definition, history)	4
Energy Systems (Phosphagen, Glycolysis, Oxidative)	7
VO ₂ Max, METs, and Measuring Intensity	6
Cardiovascular and Pulmonary Adaptations to Exercise	7
Muscle Fiber Types & Adaptation (Slow vs. Fast Twitch)	5
Exercise Prescription Principles (FITT-VP, intensity, ACSM)	7
Clinical Case: Fat Loss, Caloric Burn, RER, and Application	6
Me rambling	3



The physician crisis

- Physicians feel they are not properly educated
- 85% of US medical schools do not offer mandatory courses in physical activity or nutrition
- No behavioral counseling
- No lifestyle modification
- No preventative medicine



Physicians' Personal Health Habits

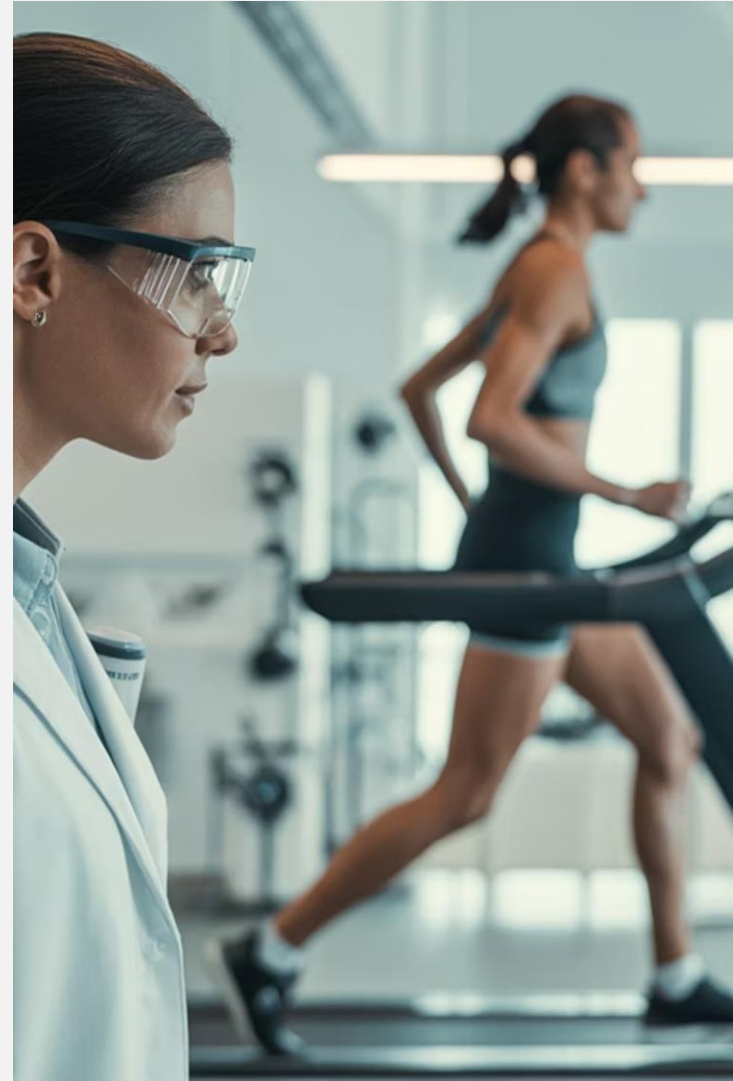
- Physically active physicians are more likely to counsel patients on healthy lifestyle and physical activity
- Physical activity habits of medical students influences their counseling after graduation



What Is Exercise Physiology?

From walking to resistance training, exercise physiology helps us understand how everyday movement impacts our bodies at the cellular, tissue, and systemic levels.

(4 slides)



Marathon Gold



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

1904 Olympic
winner, Thomas
Hicks

3:28

2022 Olympic
winner, Tamirat
Tola

2:05

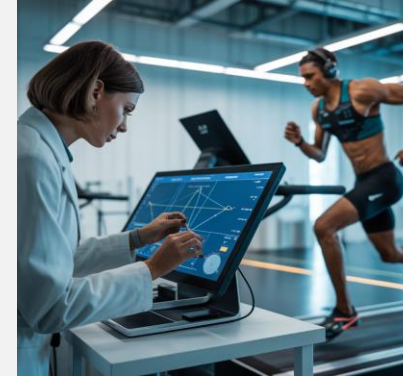
Over 1:20 minute difference



Historical Background and Scope

Sport Exercise Physiology

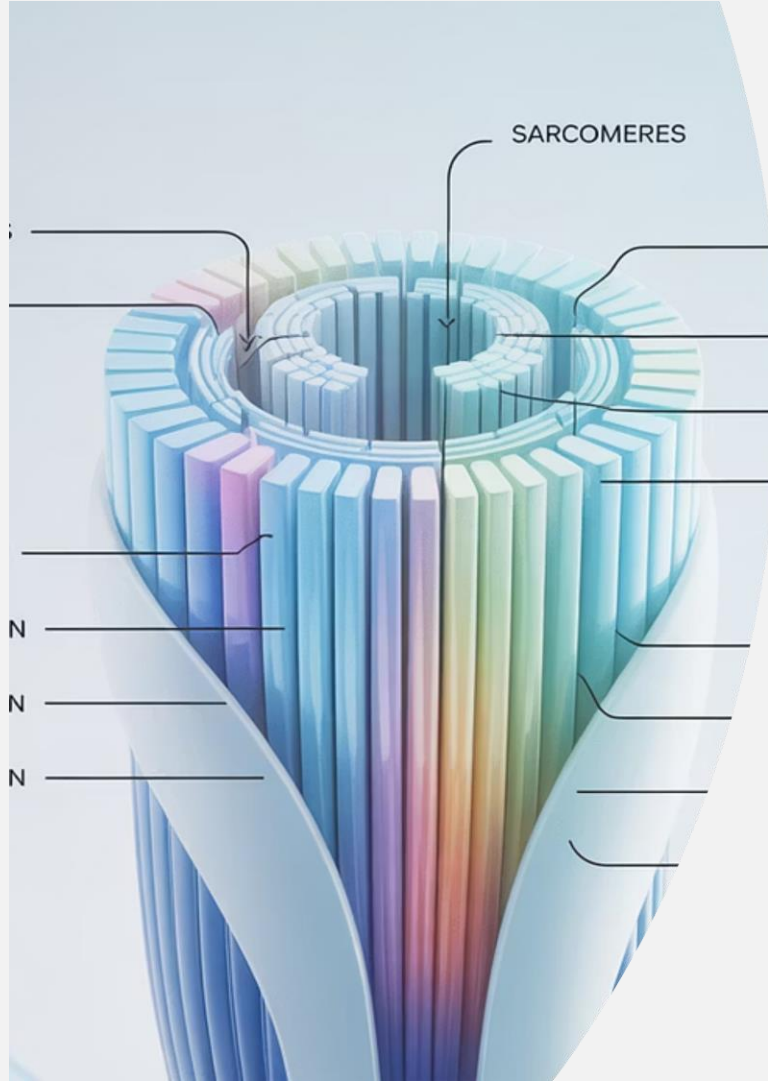
Focuses on optimizing athletic performance and training methodologies for competitive athletes and sports teams



Clinical Exercise Physiology

Emphasizes therapeutic applications for disease management, rehabilitation, and improving quality of life in clinical populations





Physiological Systems Involved

1

Musculoskeletal System

Powers movement through sarcomeres
microscopic contractile units containing actin and myosin
filaments that slide past each other during muscle
contraction, generating force and movement

2

Cardiovascular System

Increases heart rate
stroke volume
cardiac output
delivering oxygen and nutrients to working muscles while
removing metabolic waste products

3

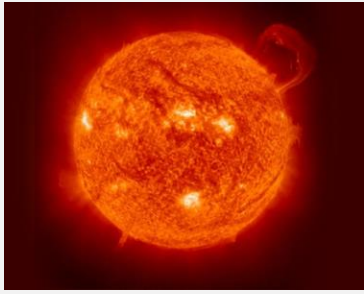
Respiratory System

Enhances oxygen uptake (VO_2) and carbon dioxide
elimination through increased ventilation rate and depth,
maintaining blood gas homeostasis during exertion

4

Nervous System

Coordinates movement patterns and activates sympathetic
responses that increase alertness, heart rate, and blood flow to
muscles while inhibiting digestive processes



- Energy



Metabolism: the total of all catabolic and anabolic reactions in a biological system.

Adenosine triphosphate (ATP)

*ATP powers all muscular activity

Energy derived from catabolic reactions occurs through an intermediate molecule.

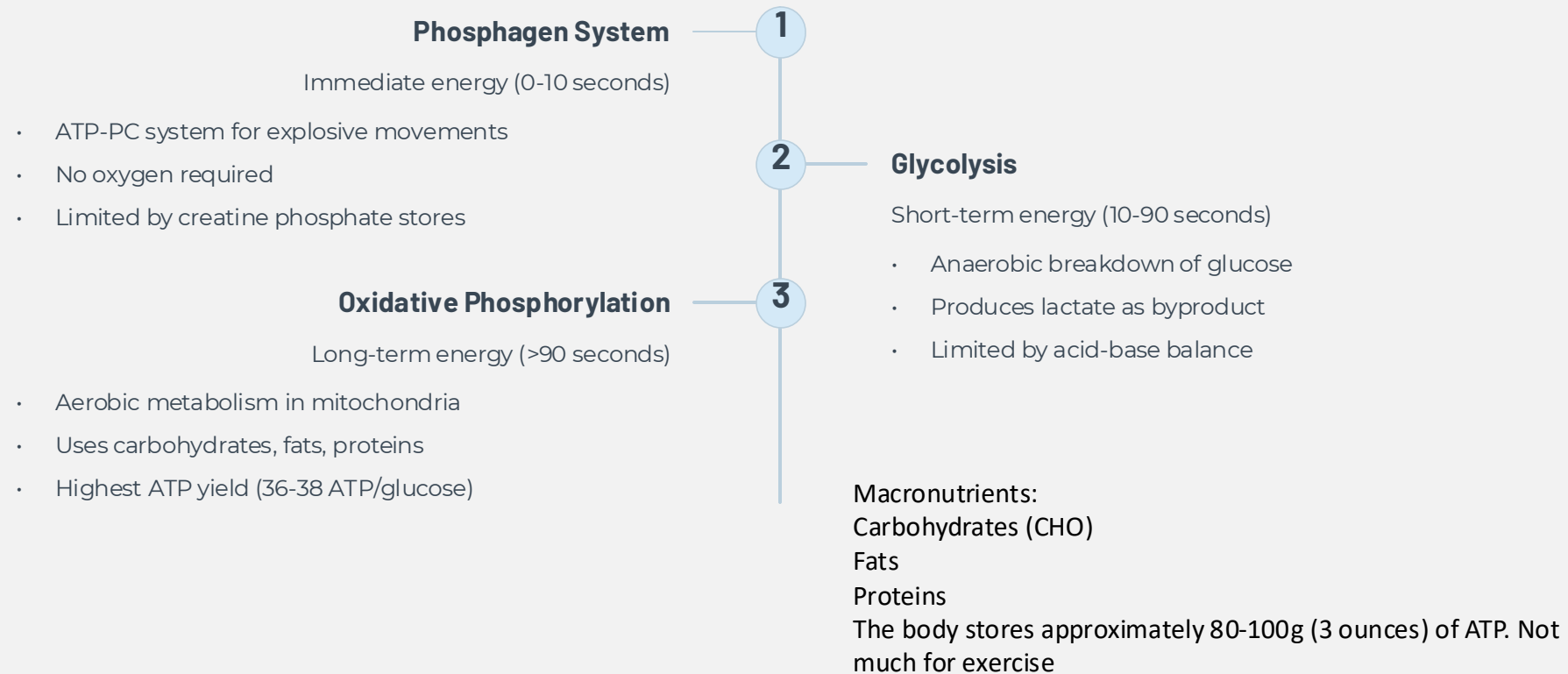
Without ATP muscular activity or growth cannot occur

So how do we produce and replenish ATP?



Energy Systems & Metabolism

How muscles acquire and utilize ATP

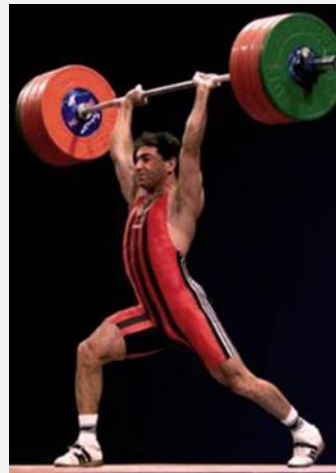


❏ **Clinical note:** Statin-induced myopathy may result from impaired mitochondrial function, directly affecting oxidative phosphorylation and reducing exercise tolerance in patients.

Phosphagen system

Type II (fast twitch fibers) contain higher levels of CP than Type I (slow-twitch)

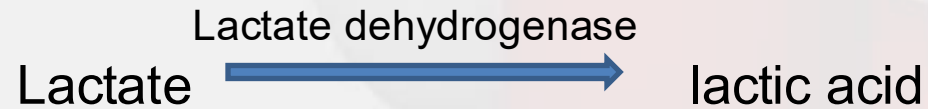
CP levels are 4x higher in skeletal muscle than ATP levels
Individuals with percentage of Type II fibers replenish ATP faster



Pyruvate

Pyruvate can be converted to lactate

Pyruvate can be shuttled into the mitochondria (slow)



Lactic acid decreases pH (approx. 7) and leads to an increase in H^+ ion concentrations inhibits glycolysis.

The ↓ in pH from exercise is called *metabolic acidosis*

Responsible for peripheral fatigue that occurs during exercise

Activity and Primary System Used

Duration of activity	Intensity of activity	Primary energy system
0-10 seconds	Extremely high	Phosphagen
10-30 seconds	Very high	Phosphagen and fast glycolysis
30 sec-2 min	High	Fast glycolysis
2-3 minutes	Moderate	Fast glycolysis and oxidative system
> 3 minutes	Low	Oxidative system



True or False:
You only burn fat in the 'fat-burning zone'
at low intensity.

While a higher ¹⁵percentage of fat is used at lower intensities, you burn more total calories — and fat — at moderate-to-high intensities. Plus, you get the EPOC (afterburn) effect

How do we measure which systems are working?



Measuring Exercise Intensity

We measure “effort” or “intensity” by

- % of maximum Heart Rate (HRR)
- Rate of Perceived Exertion (RPE)- Subjective
- Blood Lactate Threshold (anaerobic threshold)
- Ventilatory Threshold (V_e)
- **Oxygen Consumption (VO_2) (most accurate)**



Borg's Rating of Perceived Exertion (RPE) Scale

Perceived Exertion Rating	Description of Exertion
6	No exertion. Sitting & resting
7	Extremely light
8	
9	Very light
10	
11	Light
12	
13	Somewhat hard
14	
15	Hard
16	
17	Very hard
18	
19	Extremely hard
20	Maximal exertion



Maximum Oxygen Uptake (VO_2 Max)

Indirect Calorimetry

*the maximum amount of oxygen you can utilize during intense or maximal exercise.

* VO_2 max was considered the best indicator of an athlete's cardiovascular fitness and aerobic endurance.



Pulmonary response to exercise

Metabolic Cart is used to measure pulmonary function during exercise (Open-circuit spirometry)
Indirect Calorimetry



Ambient Air

O_2 - 20.93%

CO_2 - 0.03%

N - 79.04%

Changes in exhaled CO_2 and O_2 reflect ongoing metabolism in exercise

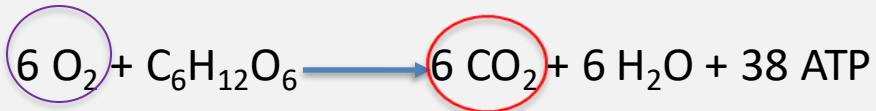
Respiratory Exchange Ratio (RER)

The ratio of CO₂ expired per minute as compared to the volume of O₂ consumed per minute.

- $RER = VCO_2 / VO_2$

RER indicates what fuel the body is using; CHO or Fat

Oxidation of a molecule of **Carbohydrate**



$$RER = VCO_2 / VO_2 = 6 CO_2 / 6 O_2 = 1.0$$

Oxidation of a molecule of **Fatty Acid**



$$RER = VCO_2 / VO_2 = 16 CO_2 / 23 O_2 = 0.7$$

What does the R mean to me?

Indirect Calorimetry

- Calorie- The energy needed to raise 1 gram of H₂O 1 degree Celsius

Indirect calorimetry Measured by $\dot{V}_E$ and exchange of CO₂ and O₂

- Pure carbohydrate catabolism 5.05 Kcal/L O₂
- Fat catabolism 4.73 Kcal/Liter O₂
- the difference in caloric expenditure between pure carbohydrate and fat catabolism, of an average healthy person exercising for 30 min at a VO₂ (oxygen consumption) of 1.5 L/min, would amount to
- $5.05 * 1.5 \text{ L/min} = 7.6 \text{ kcal/min} * 30 \text{ min} = 228 \text{ kcal CHO}$
- $4.73 * 1.5 \text{ L/min} = 7.1 \text{ kcal/min} * 30 \text{ min} = 212 \text{ kcals fat}$

Intensity	Calories Burned	% from Fat	Fat Calories
Low	200 kcal/hr	70%	140 kcal
Moderate	400 kcal/hr	50%	200 kcal

If you're trying to lose fat, caloric deficit matters more than fat as a % fuel. Moderate to high-intensity workouts can burn more total fat in less time.

What affects your $\dot{V}O_2$

Genetics and Training

- Muscle type
- Oxidative enzymes
- A- $\dot{V}O_2$ difference
- Cardiac Output (Q)

Altitude

- O_2 saturation

Mode of Exercise

- Quantity of muscle activated
- TDM or Bike?
- Swimmer on Treadmill?

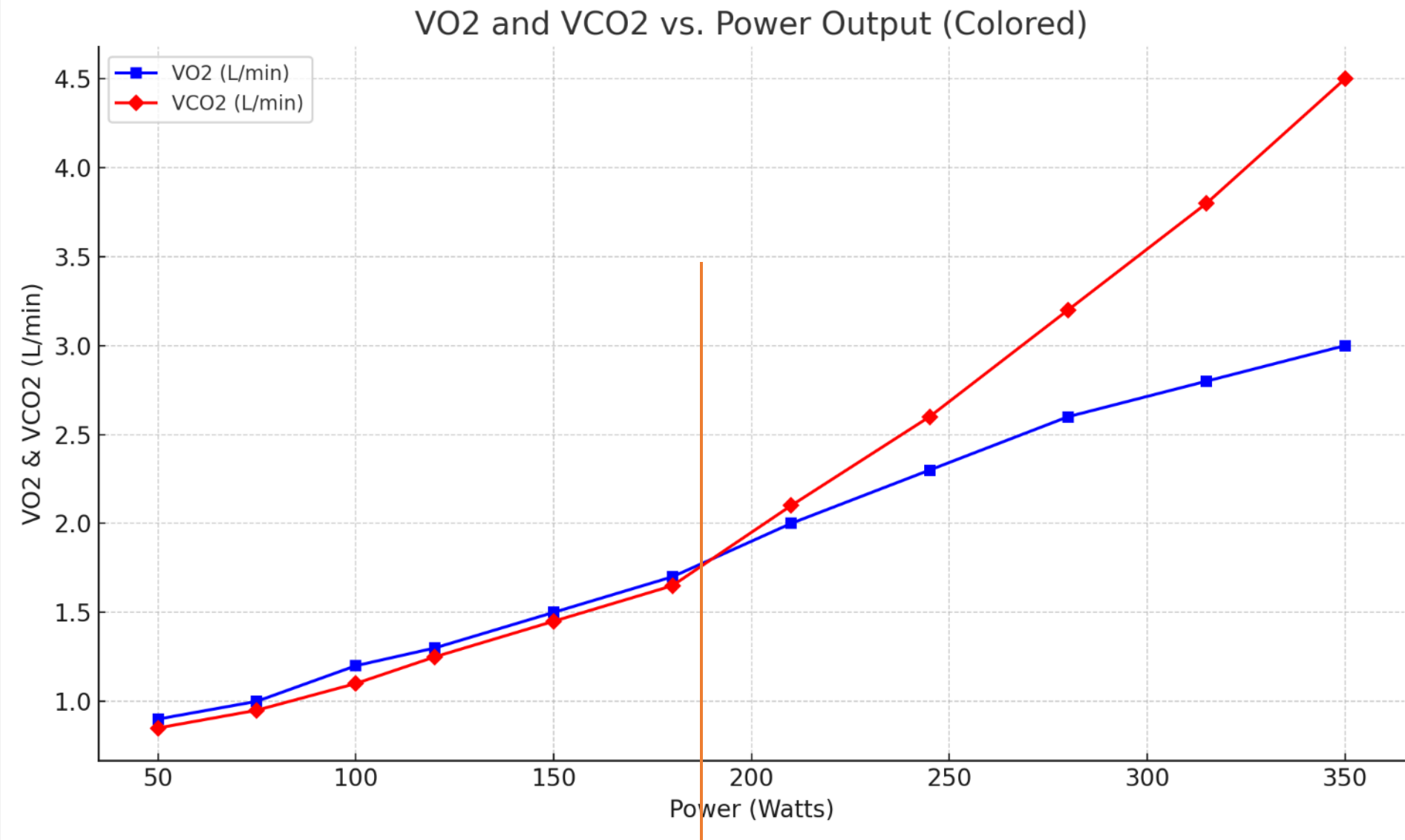
Gender

- Women tend to be lower
- Body composition
- Hemoglobin- allows for oxygen carrying ability

Age

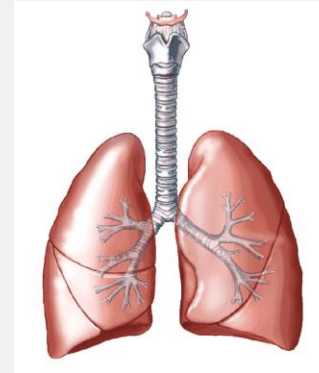
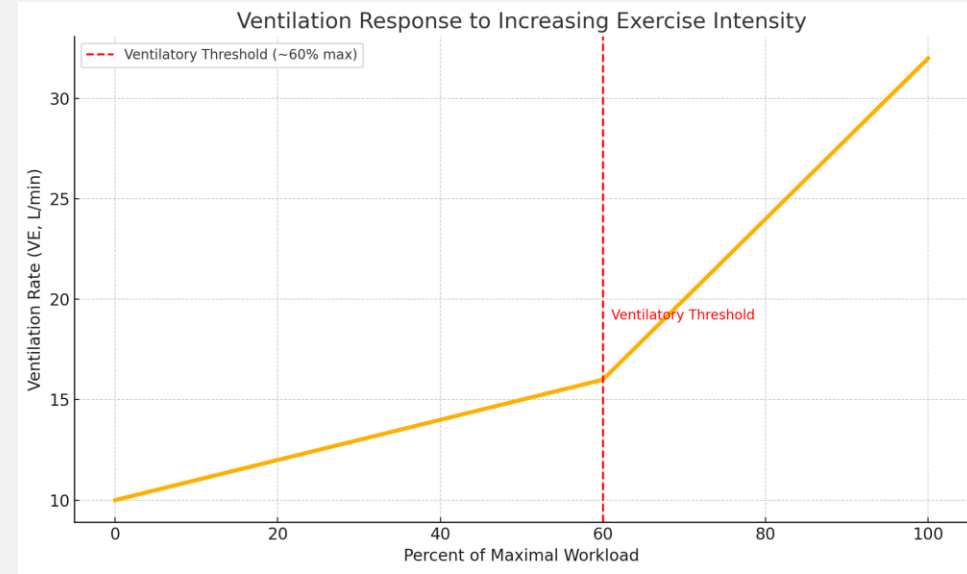
- ~ 1% per year
- people who exercise maintain it
- loss can be greater for athletes due to initial value

*Q and a- $\dot{V}O_2$ difference are main factors in changing $\dot{V}O_2$.

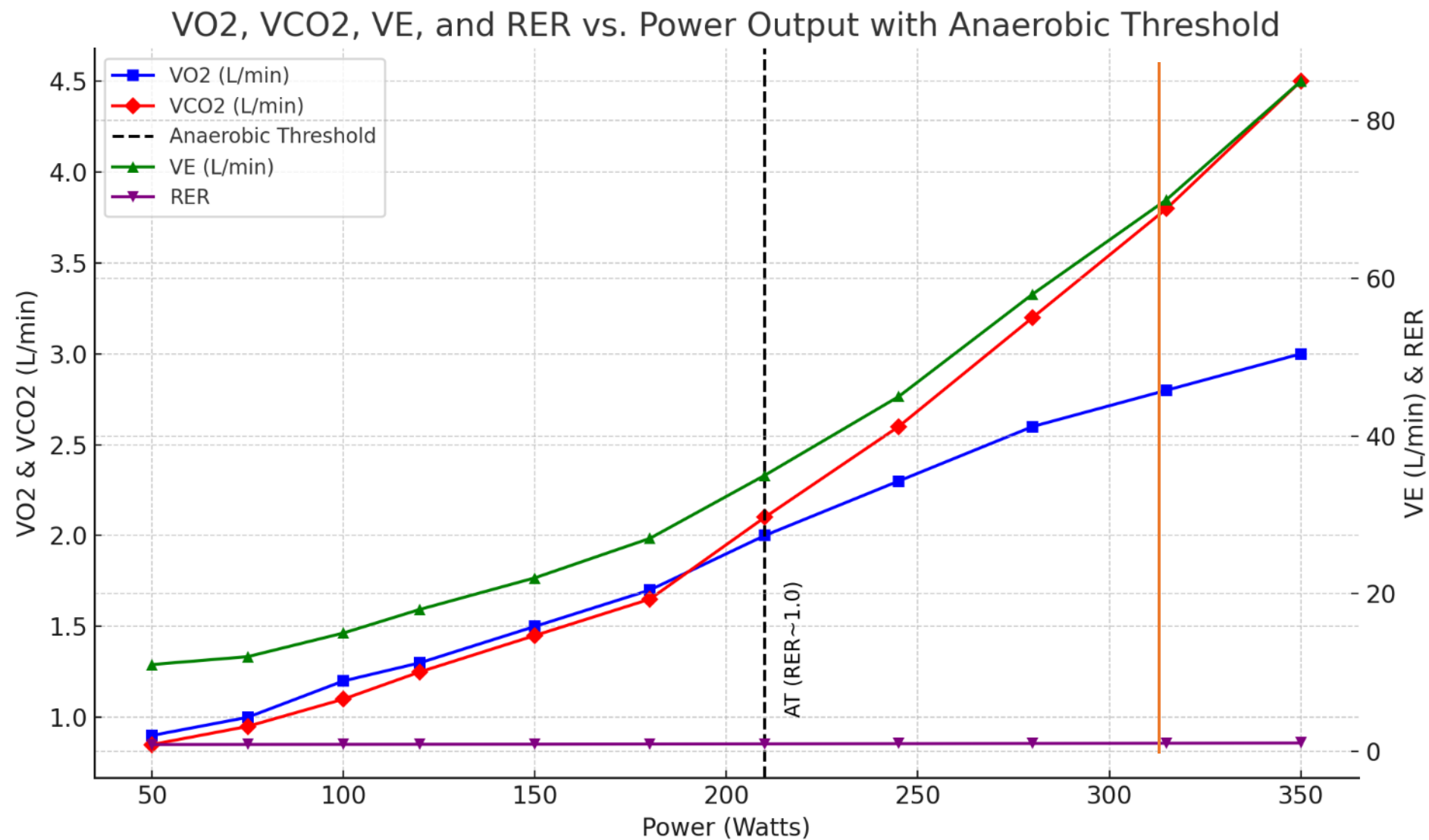


Ventilation (V_e)

- The amount of air we move in and out of our lungs
- Breaths per minute (rest 12-25)
- Increases linearly with $\dot{V}O_2$ and workload until about 60% of maximal exercise then it increases at a higher rate (40-50)



Ventilatory Threshold



Lactate Threshold/Anaerobic Threshold

* the anaerobic threshold is the better and most consistent predictor of endurance performance and sports.

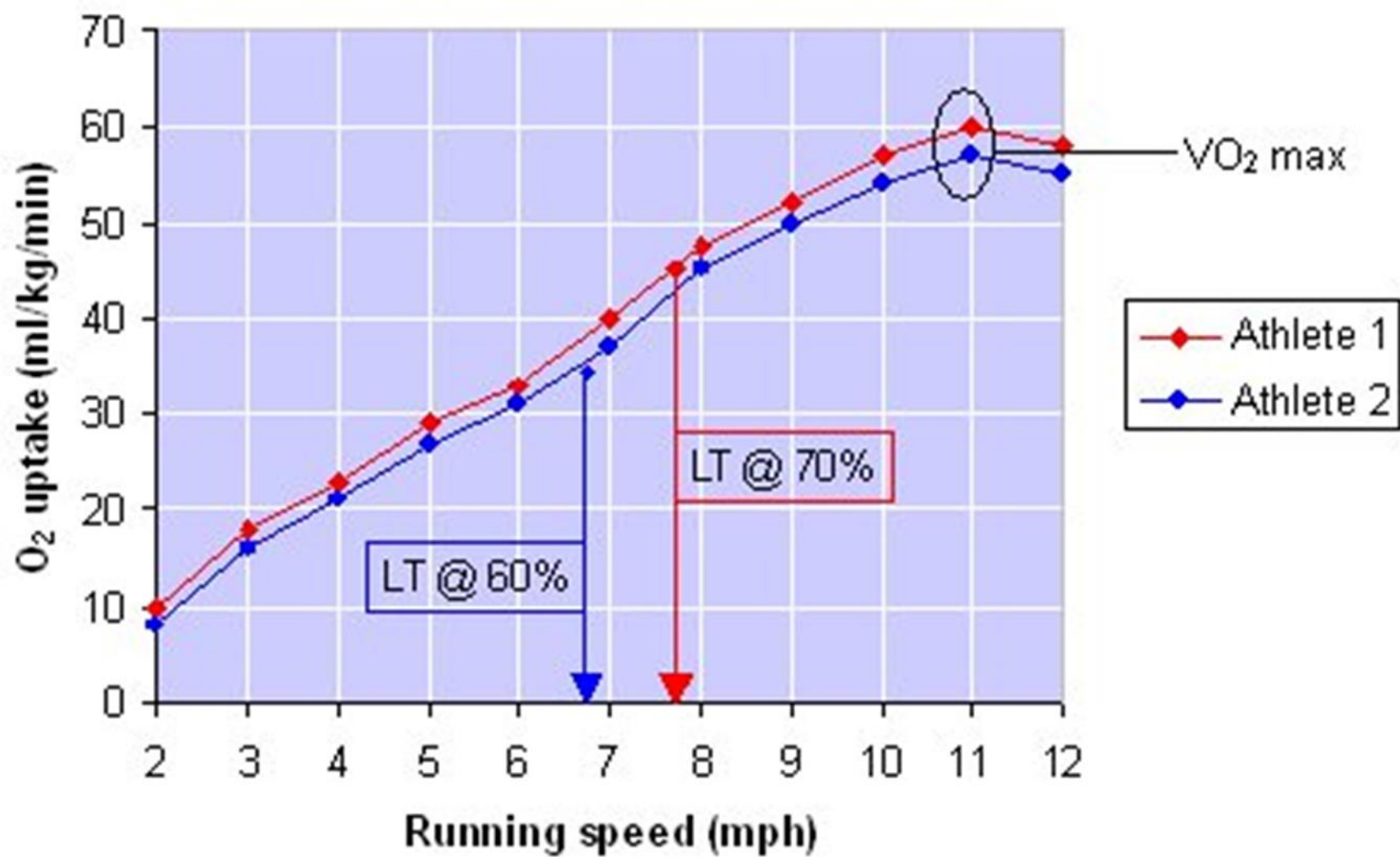
- Ventilatory Threshold
- Lactate Threshold

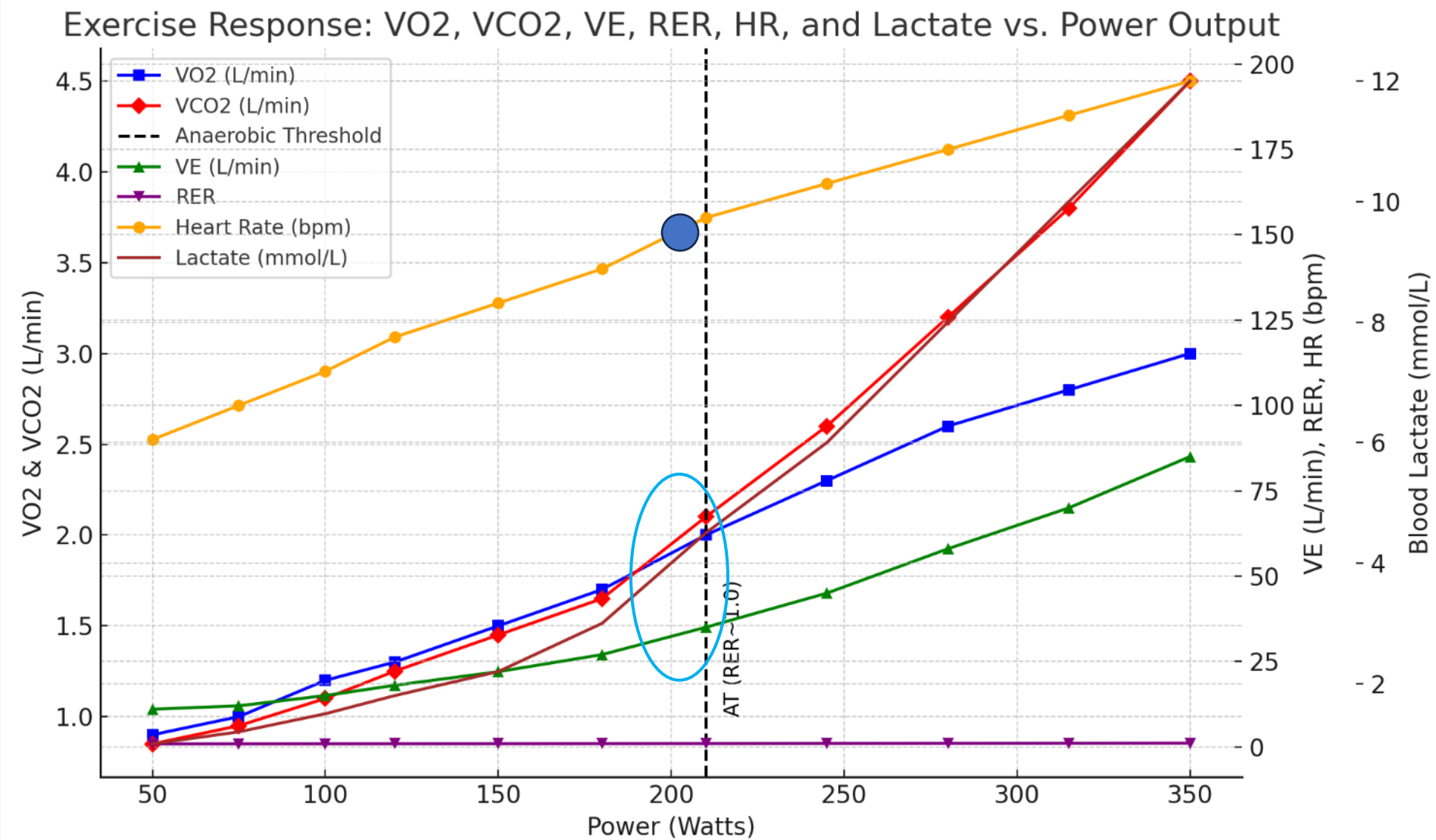
LT usually occurs at ~50–60% of VO_2max in untrained individuals, but can rise to 75–85% in trained athletes.

It can be detected via blood lactate sampling or ventilatory markers

(Wasserman, Pilegaard, Neary)



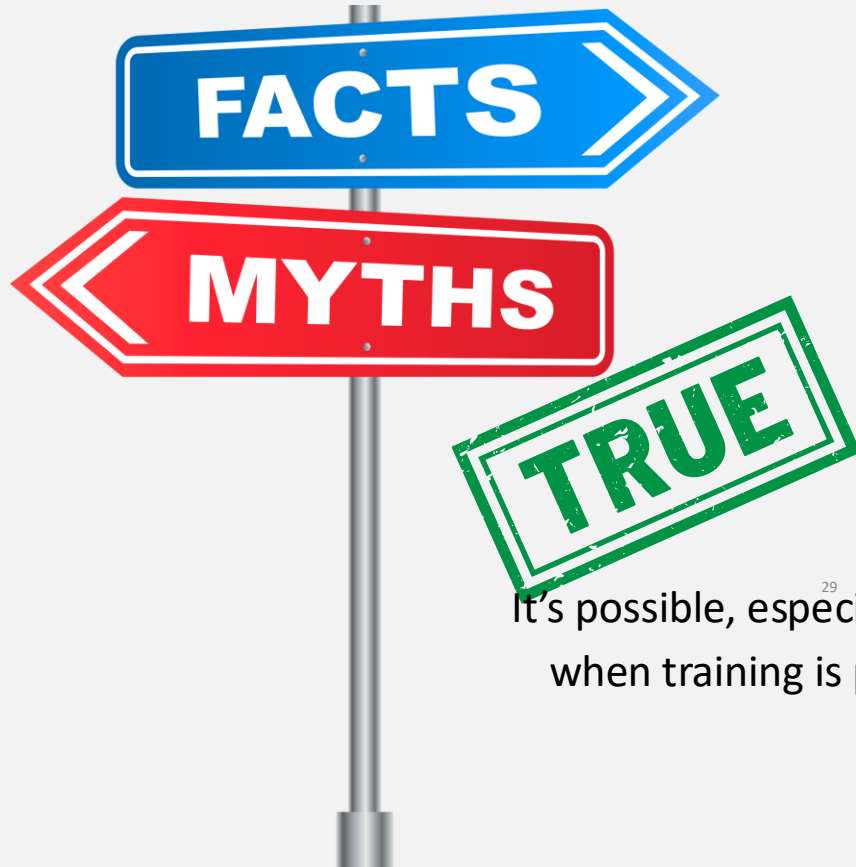




Angina
Desaturation
Dyspnea
Claudication

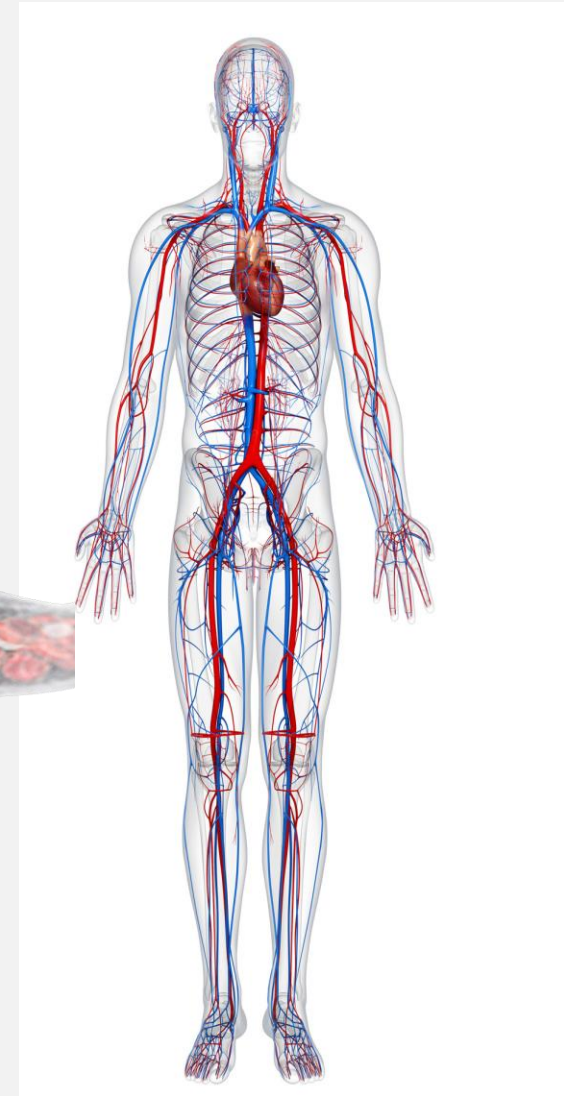
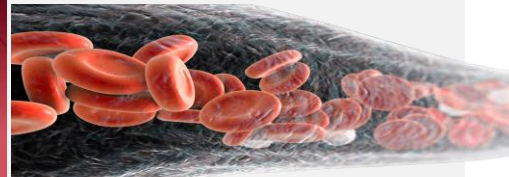
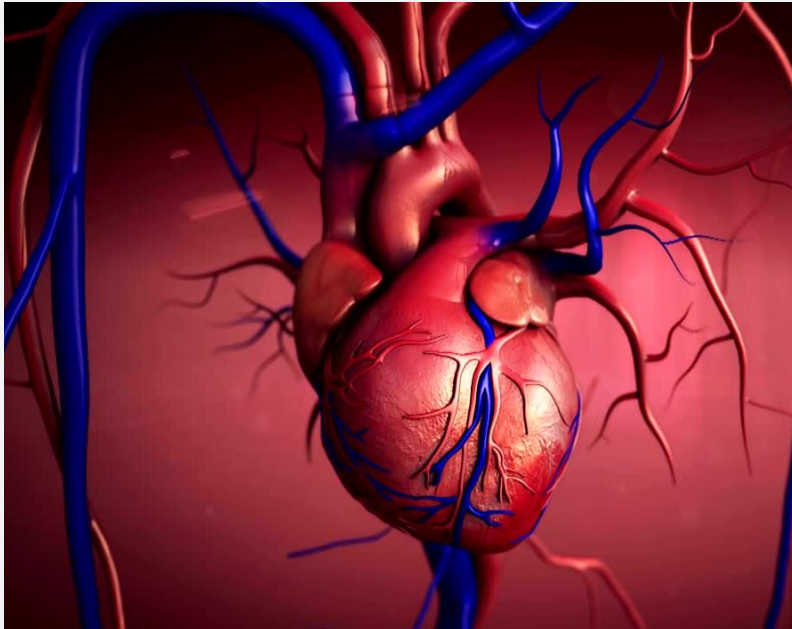
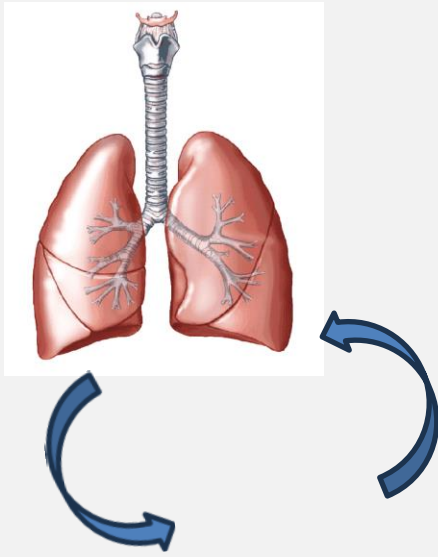
True or False:

You can't build muscle and lose fat at the same time



It's possible, especially²⁹ for beginners, those returning from a break, or when training is paired with good nutrition and adequate protein.

Cardiac Output



Cardiac Output

Cardiac Output (Q)

The amount of blood pumped by each ventricle in 1 min

Avg- 4000-5000 ml

=

Heart Rate

The number of contractions of the ventricle each minute

Avg. 70

X

Stroke Volume

The amount of blood ejected from each ventricle with each contraction

Avg. 70 ml

Factors affecting Q

=

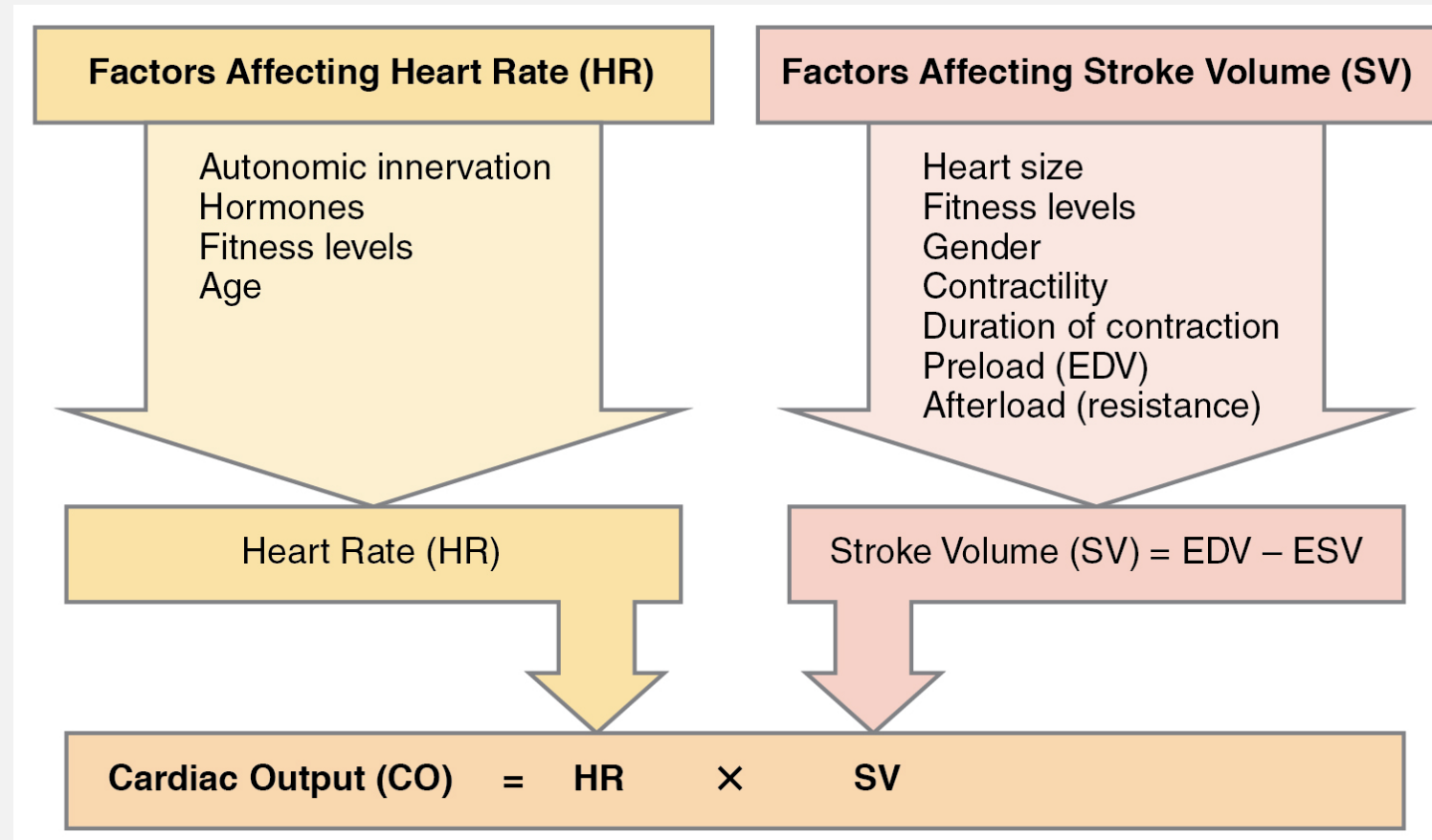
Sympathetic nervous system

epinephrine

X

- Venous return (preload)
- Blood Volume
- Sympathetic nervous system
- Peripheral Resistance (afterload)

Cardiac Response to Exercise



Concentric vs Eccentric Ventricular Hypertrophy

Concentric hypertrophy- muscle of the left side of the pumping chamber increases in size and thickness. Chamber becomes smaller.

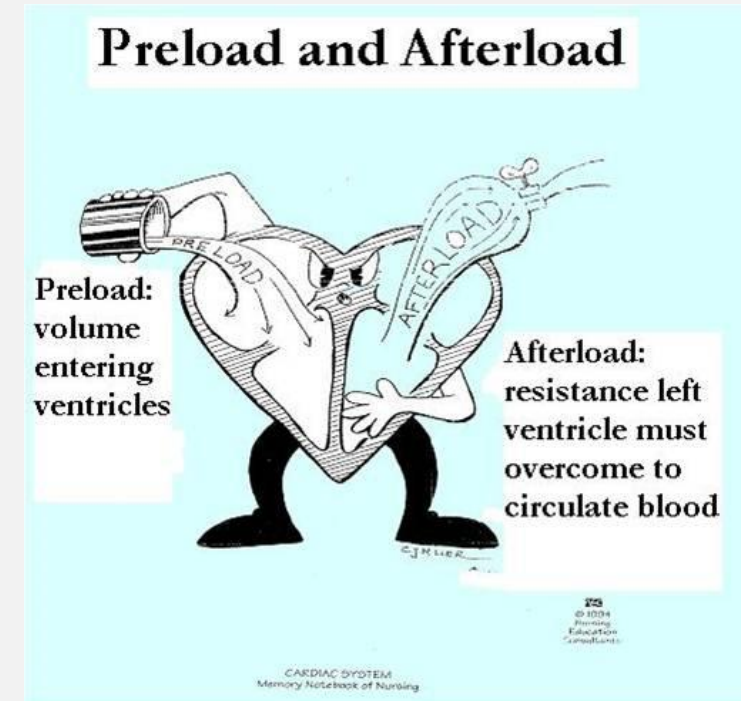
Pressure overload

↓ In stroke volume and cardiac output

- high blood pressure or any obstruction of outflow of blood from the left heart (e.g intense resistance training). Aortic valve narrowing.

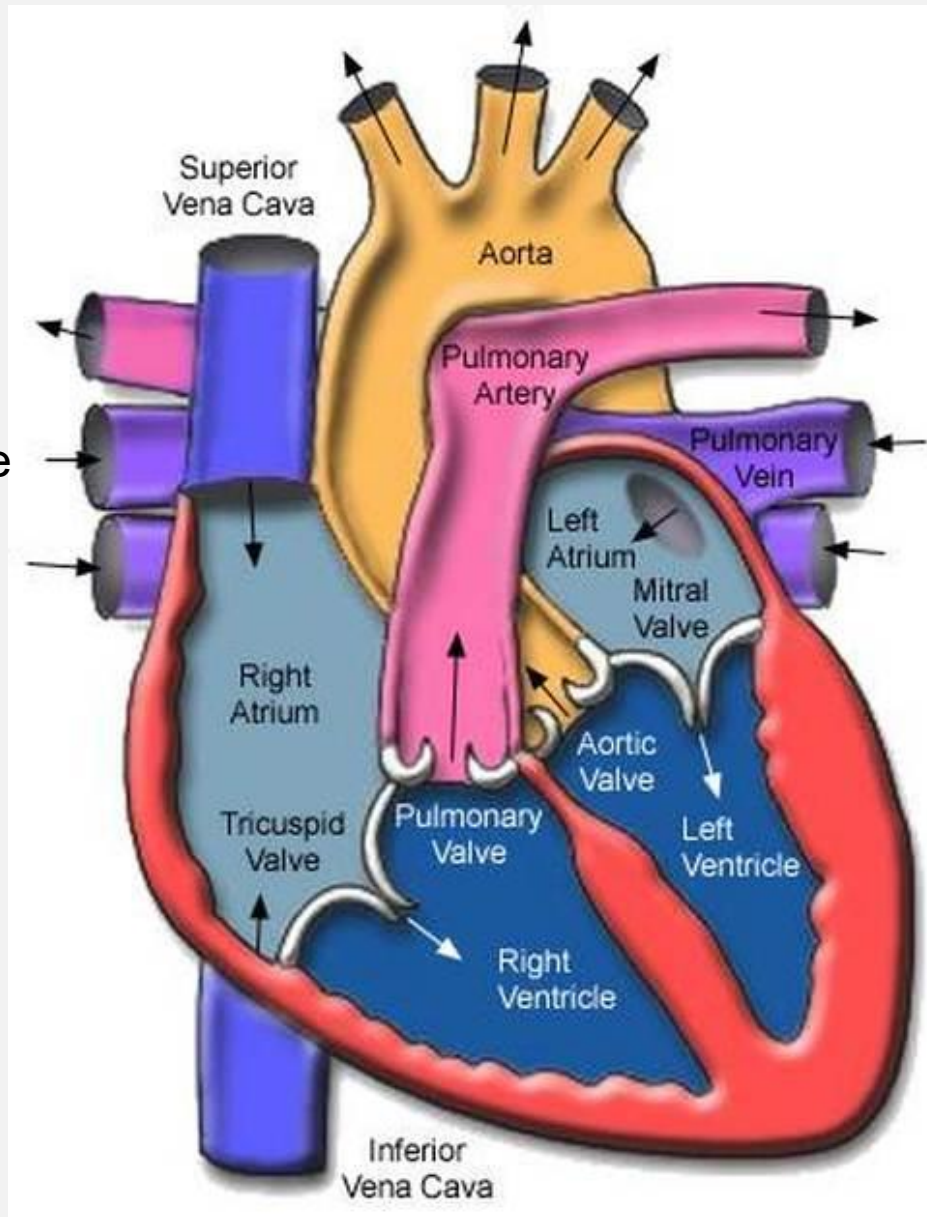
Eccentric hypertrophy- muscle of the left side of the pumping chamber thickens, increases in size and dilates. Chamber becomes larger.

- chronic preload on the heart e.g. cardiovascular exercise.
- Volume overload



Preload-volume
entering.
Increases with
cardiovascular
exercise

Increases
elasticity of left
ventricle and
chamber size



Afterload-Resistance
left ventricle must
overcome (like
squeezing a hose
during Valsalva
maneuver)
Thickens wall of left
ventricle, smaller
chamber size smaller
SV

Medical Conditions That Decrease Stroke Volume

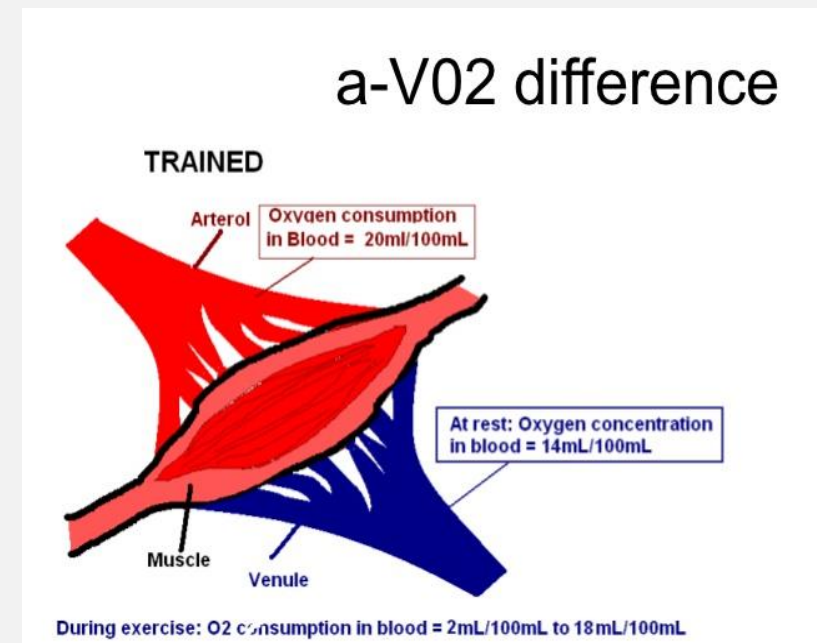
Condition	Mechanism
Heart Failure (HFrEF)	Reduced contractility, dilated ventricles
Myocardial Infarction	Damaged myocardium = impaired pump
Cardiac Tamponade	↓ Preload due to compression of the heart
Hypovolemia / Hemorrhage	↓ Preload from reduced blood volume
Pulmonary Embolism	Impaired RV output = ↓ LV preload
Severe Aortic Stenosis	↑ Afterload limits ejection
Tachycardia (severe)	↓ Filling time = ↓ Preload

Arteriovenous Oxygen Difference (A-VO₂ diff)

- The difference in O₂ content of the arterial blood and the venous blood.
- How much O₂ is diffused from the blood in capillaries and then circulated.
- Measured in mL of O₂ per 100 mL of blood (mL/100 mL)

Fick equation:

$$V\dot{O}_2 = Q \times A-V\dot{O}_2 \text{ diff}$$



Increases a-vO₂ Difference (↑ Oxygen Extraction):

Factor	Why
Exercise (especially endurance)	Muscles extract more O ₂ during increased demand
Anemia	↓ Hemoglobin → body compensates by extracting more O ₂
Sepsis (early stage)	↑ tissue metabolism → higher extraction
Heart failure (mild/moderate)	Poor perfusion = tissues compensate by extracting more
Peripheral arterial disease (PAD)	Poor delivery, so tissues extract more oxygen from what little arrives

Decreases a-vO₂ Difference (↓ Oxygen Extraction):

Factor	Why
Severe heart failure (HFrEF)	Impaired perfusion limits tissue extraction
Sepsis (late stage)	Mitochondrial dysfunction = poor oxygen utilization
Cyanide or mitochondrial toxins	Block cellular respiration → O ₂ not used despite being delivered
Shock (e.g., cardiogenic or distributive)	Tissues are underperfused and/or can't extract oxygen properly
AV fistulas or shunting	Bypasses tissue beds → less O ₂ extraction
Sedation or paralysis	↓ metabolic demand → less oxygen extracted



EXTREME TEMPERATURES

Heat and humidity are concerns for athletes competing outdoors at the Tokyo Olympics

High heat and humidity:

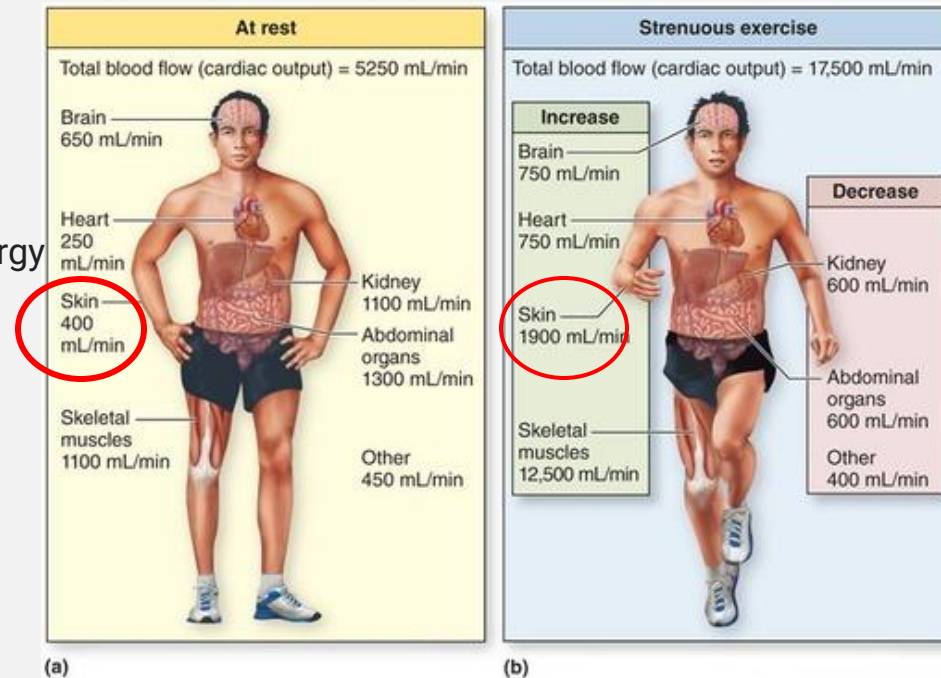
- 1) increased core body temperature
- 2) dehydration. ...

Causes decrease in VO₂max

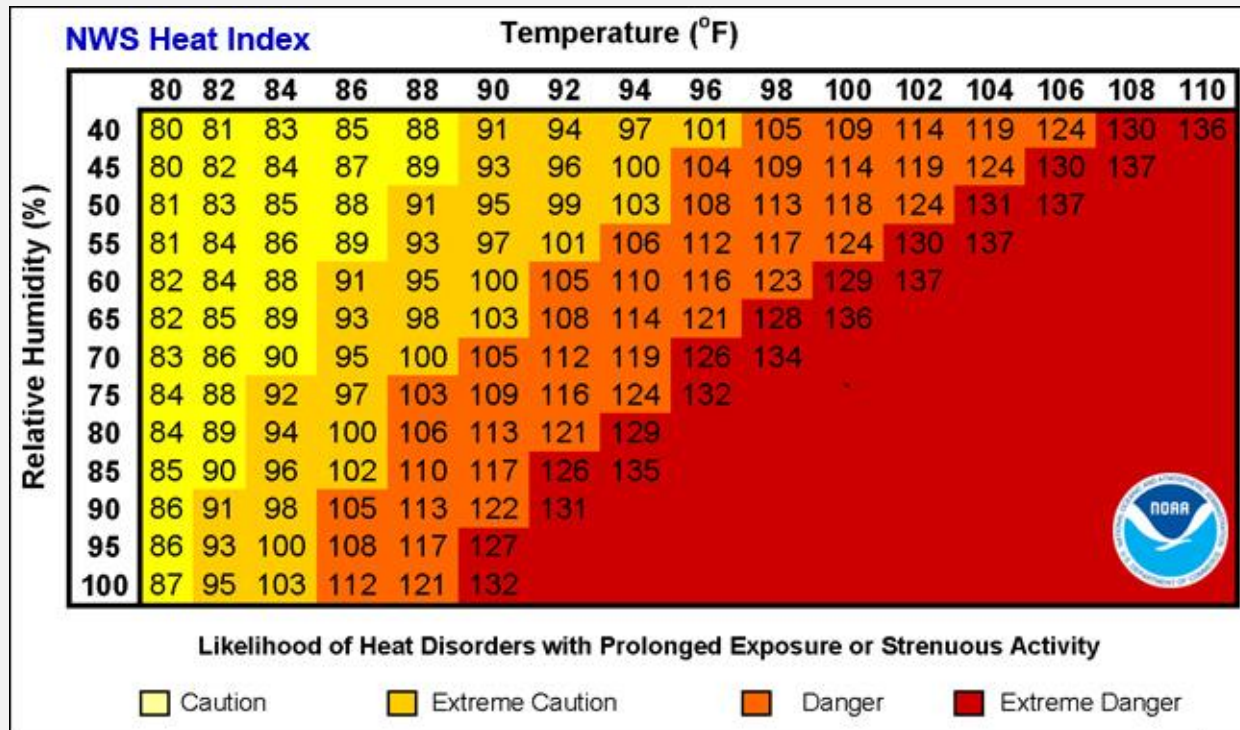
body can't utilize oxygen as efficiently to provide energy



86.9 degrees Fahrenheit (30.5 degrees Celsius) high humidity



“Its not the heat, it’s
the humidity”



Heat Index is what the
temperature feels like to the
human body

if the air temperature is
100°F and the relative
humidity is 15%, the heat
index is 96°F

Harrowing race for life as Chinese ultramarathon turns deadly in freezing weather

Yellow River Stone Forest 100km Trail Race with over 170 runners

- Top runners among 21 competitors killed in annual endurance event in Gansu



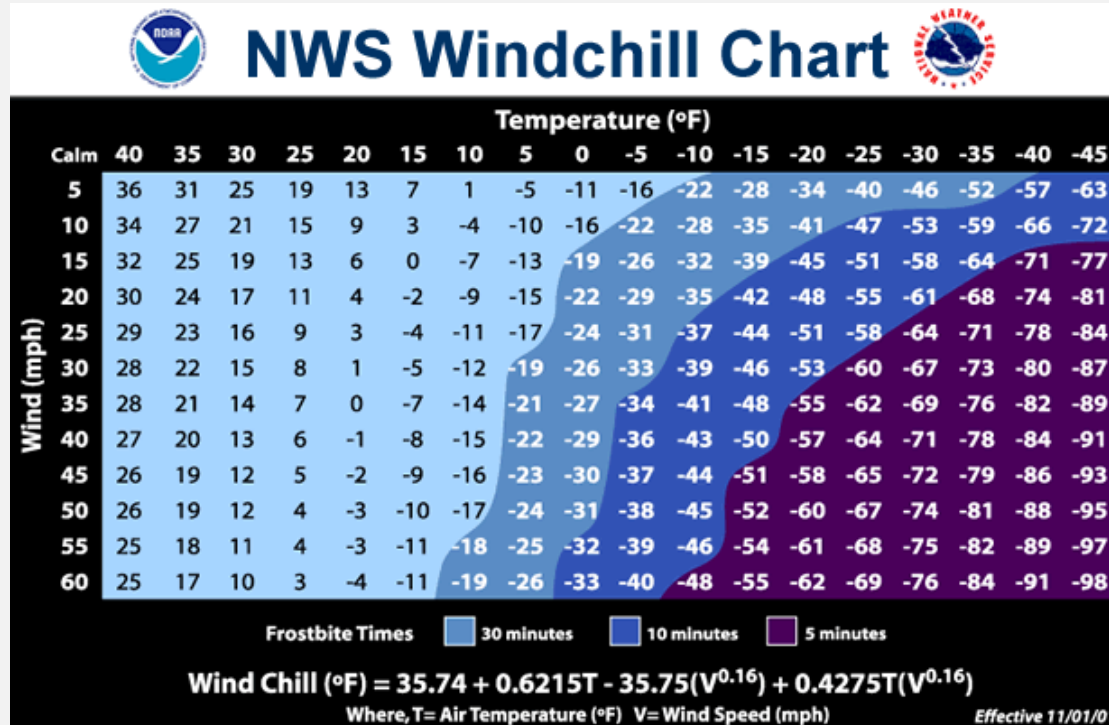
Liang Jing

-”ultramarathon tragedy was a public safety incident brought about by extreme weather including high winds, heavy rain and plunging temperatures, as well as unprofessional organization and operation”

“Twenty-one people died of hypothermia when extremely cold weather suddenly descended on a government-organized 100km (62-mile) marathon on May 22..”



When is it too cold to exercise outside?



Mayo clinic-
temperature falls below 0 degrees
Fahrenheit or the wind chill is
extreme (minus 25 degrees), you
should skip your outdoor workout

Extremely cold temps can strain on your lungs—
coughing, wheezing, and tightness in the chest

wearing a scarf over your nose and mouth to warm
up the air you're breathing may help

Frost Nip
Numbness in extremities

Why VO₂ max Decreases in Extreme Weather

Factor	Heat ☀️	Cold ❄️
Cardiac Output	Blood shunted to skin for cooling → ↓ central blood volume	Vasoconstriction ↑ afterload → ↑ cardiac strain
Muscle Oxygenation	Reduced muscle perfusion (due to skin blood flow priority)	Less oxygen delivery due to vasoconstriction
Oxygen Utilization	Enzyme efficiency drops if body overheats	Slowed metabolic reactions and nerve conduction
Hydration Status	Dehydration → ↓ plasma volume → ↓ stroke volume	Dehydration risk still present with cold diuresis
Pulmonary Function	↑ Breathing rate in heat → less efficient exchange	Cold, dry air → bronchoconstriction, discomfort
VO₂max Effect	↓ VO ₂ max due to cardiovascular and thermoregulatory strain	↓ VO ₂ max due to delivery and metabolic inefficiency

Why do cardiac events occur when people shovel snow?



Altitude

Physiological changes occur >3,900 ft

Acute

- Reduced partial pressure of O₂
- Increased cardiac output (rest/exercise)
- Increase in HR
- SV remains the same
- Increase in V_e due to increase in BR
- Body fluids become more alkaline due to ↓ in CO₂ with hyperventilation

• Longer Term

- 10-14 days HR and Q begin to return to normal due to increased red blood cell production
- Increased capillarization
- Increased mitochondria
- Increased a-VO₂ diff
- Increased use of free fatty acids sparing muscle glycogen

Revert in 1 month

More dehydrated at altitude :

- You exhale and perspire (air humidity is lower)
- Can lose up to 1 quart or more of water per day than normal

For every 300 meters (1000 ft. about) above 1500 meters elevation (about 5000ft), VO₂max decreases by 1.5 to 3.5%.

E.g.

6000ft there is about 80% of the available oxygen you get at sea level.

Adaptation

FITT-VP Principle



Exercise principles



Frequency

How often exercise is performed.



Intensity

How difficult the exercise is.



Time

How long the duration is.



Type

Mode of exercise being performed.

V- Volume P- Progression

How does
Anaerobic
Threshold and
 VO_2 max
increase?





Cardiac Adaptations: Endurance vs Strength Training

Cardiac Feature	Cardio (Endurance Training)	Strength (Resistance Training)
Primary Load	Increased Preload	Increased Afterload
Heart Chamber Size	Increased (Dilated LV)	Normal or Decreased
Wall Thickness	Normal or Slightly Increased	Increased (Thickened LV Wall)
Cardiac Remodeling Type	Eccentric Hypertrophy	Concentric Hypertrophy
Main Benefit	Increased Stroke Volume, Diastolic Function	Increased Contractile Force
Potential Concern	Rare in healthy individuals unless extreme	Reduced Compliance, Diastolic Dysfunction

Cardiovascular & Pulmonary Adaptations

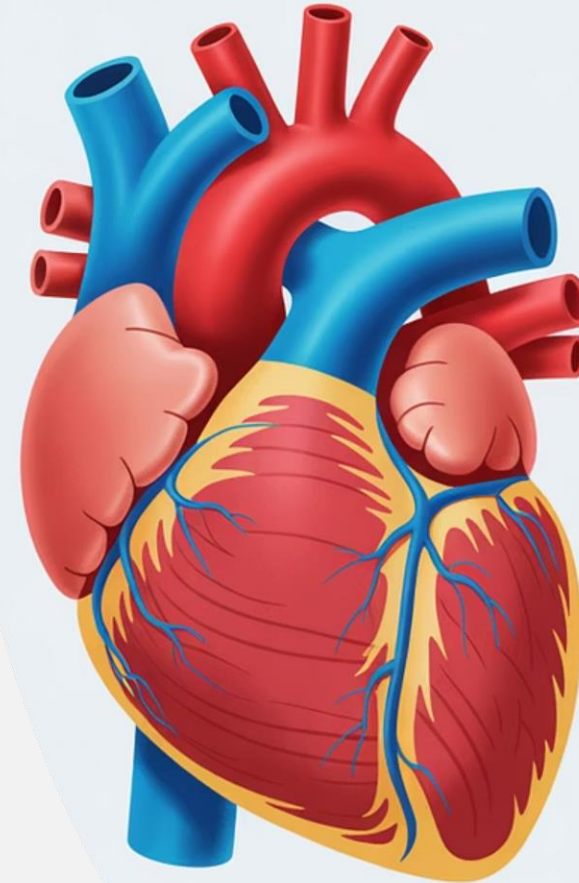
Acute Responses

- ↑ Heart rate (chronotropic)
- ↑ Stroke volume (inotropic)
- ↑ Cardiac output
- ↑ Systolic BP, ↓ Diastolic BP
- ↑ Respiratory rate and depth

Chronic Adaptations

- ↓ Resting heart rate
- ↑ Blood volume and RBC mass
- ↑ Capillary density
- ↑ VO_2 max (15-20%)
- ↑ Ventilatory efficiency

⊗ **Clinical** Patients with Heart Failure with Preserved Ejection Fraction (HFpEF) fatigue quickly due to impaired diastolic filling and reduced stroke volume augmentation during exercise.



How Fast Do We Adapt to Exercise?

Exercise triggers rapid and layered physiological adaptations:

Physiological Adaptations to Exercise Over Time

Plasma volume
increase

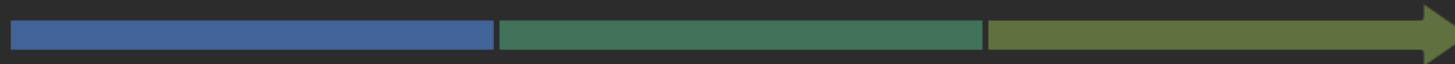
7–10 days

Mitochondrial
density and
capillarization
increase

3–4 weeks

VO₂ max
improvements

4–6 weeks

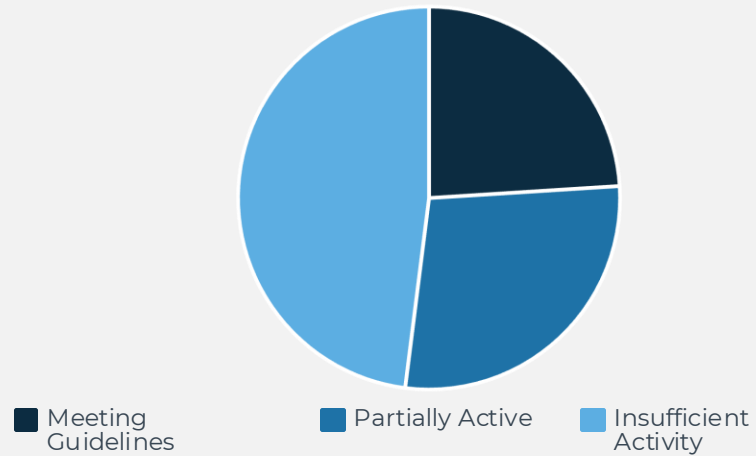


Physiological Adaptations to Exercise Over Time



Neuromuscular adapts fastest, cardio next,
muscular growth takes time

Challenges and Considerations



48% of adults worldwide do not meet recommended physical activity guidelines.

Healthcare providers face several barriers to effective exercise prescription, including time constraints during consultations, insufficient training in exercise science, and lack of reimbursement for counseling.

The most successful exercise interventions focus on enjoyable, sustainable activities that patients can realistically incorporate into their daily routines, with clear progression plans and regular follow-up.

Summary

1. The body uses three systems to generate ATP for movement: phosphagen, glycolytic, and oxidative.
2. Exercise intensity is the key factor that determines which energy system is dominant—not just duration or frequency.
3. VO_2 is influenced by multiple factors, including cardiac output, oxygen delivery, and muscle utilization.
4. Medical conditions (e.g., heart failure, COPD, neurological disorders), medications (e.g., beta blockers, vasodilators), and environmental stressors (e.g., heat, cold, altitude) can all alter normal physiological responses to exercise.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

Clinical Scenario:

Patient:

32-year-old female, BMI 32, sedentary lifestyle

Complaint:

“I want to lose weight, mostly body fat. I’ve been told to keep my heart rate low to burn fat, but I don’t know if that’s enough.”

Vitals:

- Resting HR: 78 bpm
- BP: 118/76
- No known medical conditions
- Normal labs, no diabetes or thyroid issues

Current activity:

Walks 20 minutes 3x/week at a casual pace (approx. 50% max HR)

Based on her goal of fat loss and current fitness level, should you recommend she continue low-intensity walking or increase to moderate-to-high intensity exercise?

Use knowledge of energy systems, fat vs. carbohydrate utilization, and total caloric expenditure to support your recommendation.

Which of the following is the **most appropriate exercise recommendation** to support this patient's goal of **fat loss**?

- A.** Continue low-intensity walking to stay in the “fat-burning zone” since fat is used more at lower intensities.
- B.** Increase to moderate-intensity exercise (60–75% HRmax) to enhance total caloric expenditure and improve fitness.
- C.** Switch to high-intensity interval training (HIIT) only, since low-intensity exercise does not burn fat.
- D.** Avoid increasing intensity until weight loss begins, as moderate or high-intensity exercise could promote carbohydrate use and limit fat burning.

Correct Answer: B

 **Explanation:**

- While it's true that a higher percentage of fat is burned at low intensities (A), total caloric burn is much lower.
- Moderate intensity increases total energy expenditure, includes both fat and carb oxidation, improves cardiorespiratory fitness, and boosts post-exercise fat oxidation.
- Option C is too aggressive for a sedentary individual and may not be sustainable early on.
- Option D reflects a misunderstanding; weight loss is primarily driven by total energy expenditure, not just substrate preference during activity.

Case Integration: Clinical Application

"45-year-old sedentary male starts jogging, experiences rapid fatigue and lightheadedness"

1

Limiting Systems

- Underdeveloped oxidative energy system
- Poor cardiovascular conditioning (↓ stroke volume)
- Inefficient ventilatory response
- Low lactate threshold

2

Expected 4-Week Adaptations

- 10-15% increase in VO_2 max
- Improved stroke volume and cardiac output
- Enhanced mitochondrial density
- Increased lactate threshold
- Delayed onset of fatigue and symptoms

When prescribing exercise, remember that physiological adaptations occur across multiple systems simultaneously, creating a coordinated improvement in functional capacity.

FROM CPG Indications to exercise test

- Diagnose coronary artery disease
- Diagnose chest pain, shortness of breath or lightheadedness
- Determine a safe level of exercise
- Check the effectiveness of procedures done to improve coronary artery circulation in patients with coronary artery disease
- Predict risk of dangerous heart-related conditions such as a heart attack.
- Depending on the results of the exercise stress test, the physician may recommend more tests such as a nuclear stress test or cardiac catheterization

Indications for Exercise Testing Pulmonary Patients

- Cause of SOB remains unclear despite PFT's
- To distinguish whether cardiac or respiratory disease limits exercise tolerance
- To evaluate for exercise –induced O_2 desaturation
- To determine appropriate intensity for exercise training

Normal Exercise Responses

ACSM Guidelines 8th edition

Variable	Response	Clinical Significance
Heart Rate	Linear ↑ with increase intensity ~10 beats per MET	Reduced with Beta blockers, and certain medications
Systolic BP	Linear ↑ ~10 Hg/MET. May peak at max exercise	>250 mm Hg or hypotension > 10 mm Hg may indicate myocardial ischemia / LV dysfunction
Diastolic BP	No change or slight decrease	End test > 15 mm Hg > 115 mm Hg
V _O ₂	Linear ↑ with ↑ intensity	Will plateau and V _e will rise
O ₂ saturation	A ↓ of 4% is considered abnormal	May indicate respiratory disease
ST wave changes	There should be no upward or downward sloping	≥ 1.0 mm downslope Suggests myocardial ischemia ST- elevation may reflect MI aneurysm or wall motion abnormalities

Relative Contraindications for physical activity/ exercise

- Left main coronary stenosis
- Moderate stenotic heart disease
- Electrolyte abnormalities (e.g. hypokalemia, hypomagnesemia)
- Severe arterial hypertension (i.e. systolic BP of >200mm Hg and/or a
- diastolic of BP of >110mm Hg) at rest
- Tachydysrhythmia or bradydysrhythmia
- Hypertrophic cardiomyopathy and other forms of outflow tract obstruction
- Neuromuscular, musculoskeletal, or rheumatoid disorders that are
- exacerbated by exercise
- High-degree atrioventricular block
- Ventricular aneurysm
- Uncontrolled metabolic disease (e.g., diabetes, thyrotoxicosis, or myxedema)
- Chronic infectious disease (e.g. mononucleosis, hepatitis, AIDS)
- Mental or physical impairment leading to inability to exercise adequately

ACSM Guidelines for exercise Testing and Prescription 8th ed.

American Heart Association Guidelines for exercise testing http://www.heart.org/HEARTORG/Conditions/HeartAttack/DiagnosingaHeartAttack/Exercise-Stress-Test_UCM_307474_Article.jsp#.V7tGWk0rLcs

Absolute termination guidelines to exercise testing

- A recent significant change in the resting ECG suggesting significant
- ischemia, recent myocardial infarction (within 2 days) or other acute
- cardiac event
- Unstable angina
- Uncontrolled cardiac dysrhythmias causing symptoms or hemodynamic
- compromise
- Symptomatic severe aortic stenosis
- Uncontrolled symptomatic heart failure
- Acute pulmonary embolus or pulmonary infarction
- Acute myocarditis or pericarditis
- Suspected or known dissecting aneurysm
- Acute systematic infection